Dhairya Patel

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Education

VIT Bhopal University 2023 – 2027
B.Tech - Computer Science and Engineering — CGPA: 8.40 *Madhya Pradesh*

Skills

Technical Skills

Languages: Python, C++, SQL
Frameworks & Libraries: NumPy, Pandas, Matplotlib, yfinance, Scikit Learn, TensorFlow
Developer Tools: VS Code, Jupyter

Quantitative Skills

Derivatives, Futures Pricing, Hedging Strategies, Portfolio Optimization (In Progress)

Projects

Reinforcement Learning-Based Stock Trading Simulator | *Python, NumPy Matplotlib* Personal Project

- Developed a **Q-learning based trading agent** to dynamically optimize portfolio allocation in a simulated stock market environment.
- Modeled asset prices using **Geometric Brownian Motion** and engineered custom **state-action spaces** with an epsilon-greedy exploration strategy.
- Evaluated performance against a **buy-and-hold benchmark**, achieving **improved risk-adjusted returns** across multiple simulated price paths and highlighting learning limitations under market uncertainty.

Custom Auto Sector Index Construction | *Excel, Market Analysis, Financial Modeling* Personal Project

- Constructed a **9-stock custom Auto sector index** using price-weighted, market-cap weighted, free-float market-cap weighted, and equal-weighted methodologies.
- Computed index levels using **real market prices** and simulated short-term price movements to analyze index sensitivity and behavior.
- Analyzed concentration risk across weighting schemes, demonstrating how **heavyweight stocks (Maruti Suzuki, M&M)** disproportionately drive cap-weighted index volatility and returns.

Stock Wallet : Financial Advisory Website | *HTML, Tailwind CSS, JavaScript, Chart.js* Group Project

- Developed a **financial advisory web platform** generating personalized investment allocations based on user risk profile, goals, and capital.
- Implemented **SIP and lump-sum return calculators** to project portfolio growth and final corpus value over time.
- Built **interactive asset-allocation and growth visualizations** using Chart.js to enhance clarity of investment decisions.

Relevant Coursework

Computer Science: Data Structures and Algorithms, Operating Systems, Computer Architecture and Organization, Computer Networks, Database Management Systems

Mathematics: Applied Linear Algebra, Probability and Statistics, Calculus, Numerical Methods, Laplace Transform

Data & ML: Foundations of Data Science, Fundamentals of Machine Learning

Certifications

- **NISM Series VIII – Equity Derivatives (In Progress)**
- **NPTEL - Machine Learning**
- **MATLAB Onramp**
- **AWS Cloud Computing**